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FULL CRITICAL REVIEW

Six decades of the Hall–Petch effect – a survey of grain-size strengthening studies on pure metals

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Refining a metal's grain size can result in dramatic increases in strength, and the magnitude of this strengthening increment can be estimated using the Hall–Petch equation. Since the Hall–Petch equation was proposed, there have been many experimental studies supporting its applicability to pure metals, intermetallics and multi-phase alloys. In this article, we gather the grain-size strengthening data from the Hall–Petch studies on pure metals and use this aggregated data to calculate best estimates of these metals' Hall–Petch parameters. We also use this aggregated data to re-evaluate the various models developed to physically support the Hall–Petch scaling.

Keywords: Hall–Petch, Grain-size strengthening, Nanocrystalline metals, Dislocations

Introduction

In the early 1950's, Hall[1] and Petch[2] empirically demonstrated that the initial yield point of low carbon steels, σ_y, was related to their grain size, D, according to the now well-known Hall–Petch relationship:

$$\sigma_y = \sigma_0 + k\frac{1}{\sqrt{D}} \tag{1}$$

where σ_0 and k are chemistry- and microstructure-dependent constants. In the six decades since Hall's and Petch's work, subsequent experimental studies have demonstrated that equation (1) or similar power laws apply not just to the initial yield of mild steels, but to the yield and flow stresses of other pure metals and alloys.[3] Despite the simplicity of equation (1) and its neglect of, e.g., crystallographic texture, other defect structures beyond grain boundaries, etc., it has proven remarkably relevant to metallurgy over that time. Because of such studies, grain-size strengthening data exists for most of the transition metals[4,5] as well as pure Be,[6] Mg[7] and Al.[8] And owing to the interest in nanocrystalline metals over the past two decades, for several of these metals there is now grain-size strengthening data spanning six decades of grain size, from millimetre to nanometer-size grains.

In this article, we gather and assess grain-size strengthening data for pure metals. Our two main motivations for this undertaking were the following. First, in the course of our own work on the design of nanocrystalline alloys, we felt the need for a reference compilation that could be used to estimate grain-size strengthening effects in pure metals and alloys, and which could be more broadly useful to the mechanics of materials and microstructural design communities. Second, we wanted to encourage renewed discussion of the possible mechanisms that give rise to power-law grain-size strengthening such as equation (1) by comparing this aggregated grain-size strengthening data with predictions from the many models that have been proposed to physically support equation (1).

Along these lines, this survey contains two main sections which address each of these goals in turn. In the first section, we present all of the currently available grain-size strengthening data we have found in the open literature, and summarize best estimates of σ_0 and k for each of the pure metals that have been studied. Subsequently, we compare the trends seen in this aggregated grain-size strengthening data with those predicted by the various theories explaining equation (1), starting with the pile-up model proposed by Hall in his original paper.[1,9] We also briefly discuss the breakdown in the Hall–Petch scaling seen in the finest grain-size nanocrystalline samples. The collection of many studies in a single analysis permits us to illustrate some effects of crystal structure, bond strength, temperature and plastic strain on σ_0 and k that have not always been easily appreciated from individual studies.

Grain-size strengthening data

We summarize the BCC transition metals' grain-size strengthening data in Fig. 1*a–g*. In each of these figures, we plot results from the various Hall–Petch studies on the different metals as well as best fits to the entirety of the data using equation (1). The open points indicate flow stresses measured using tension and compression tests while the closed points indicate Vickers and nanoindentation hardness measurements that were divided by a Tabor factor of 3 to convert to strength measurements. Figures 2 and 3 are formatted in the same way as Fig. 1, but show data for the FCC and HCP metals, respectively. Most of the flow stresses included in Figs. 1–3 were measured at plastic strains of order 0.2%, though some of the data points are from larger plastic strains, of order

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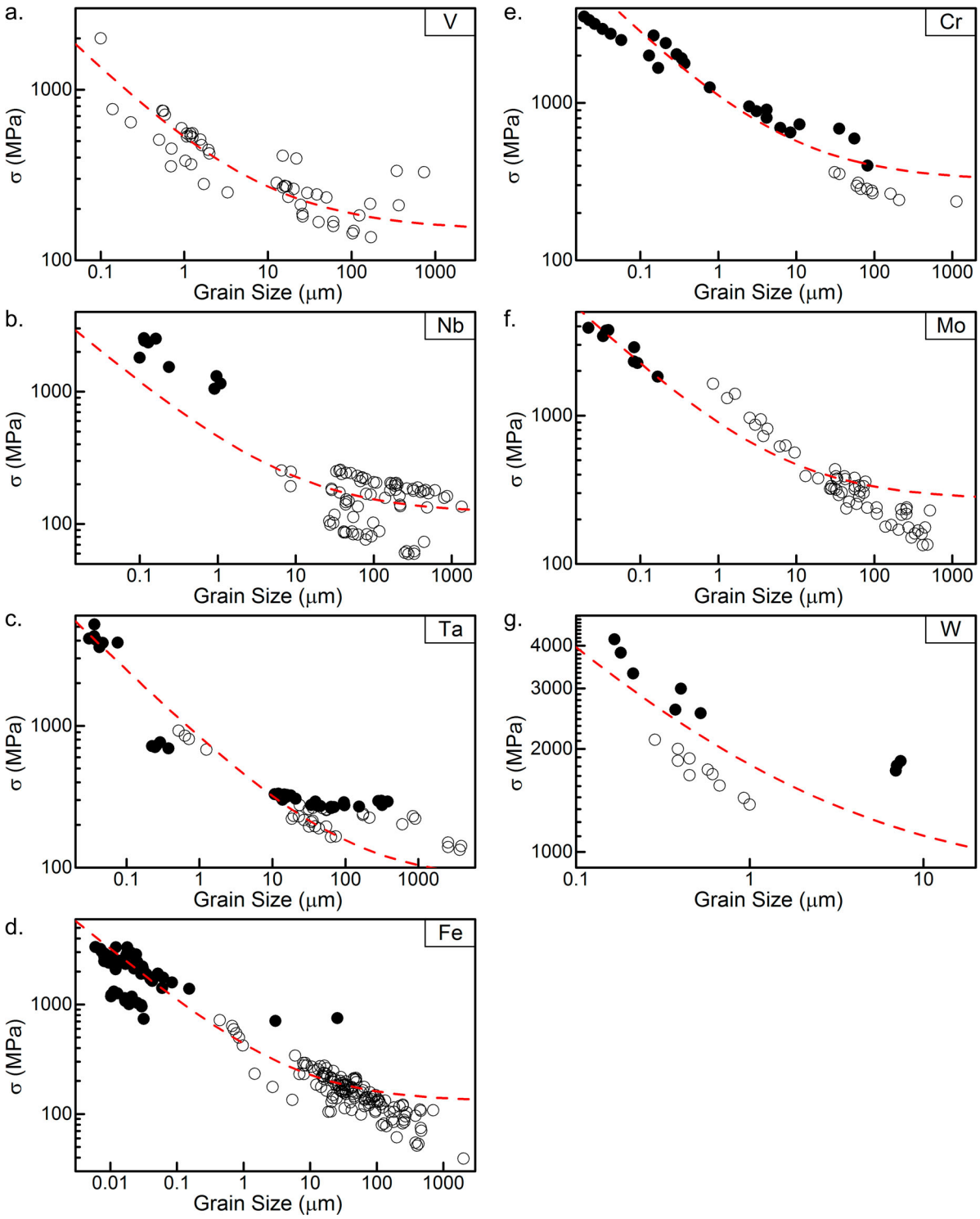
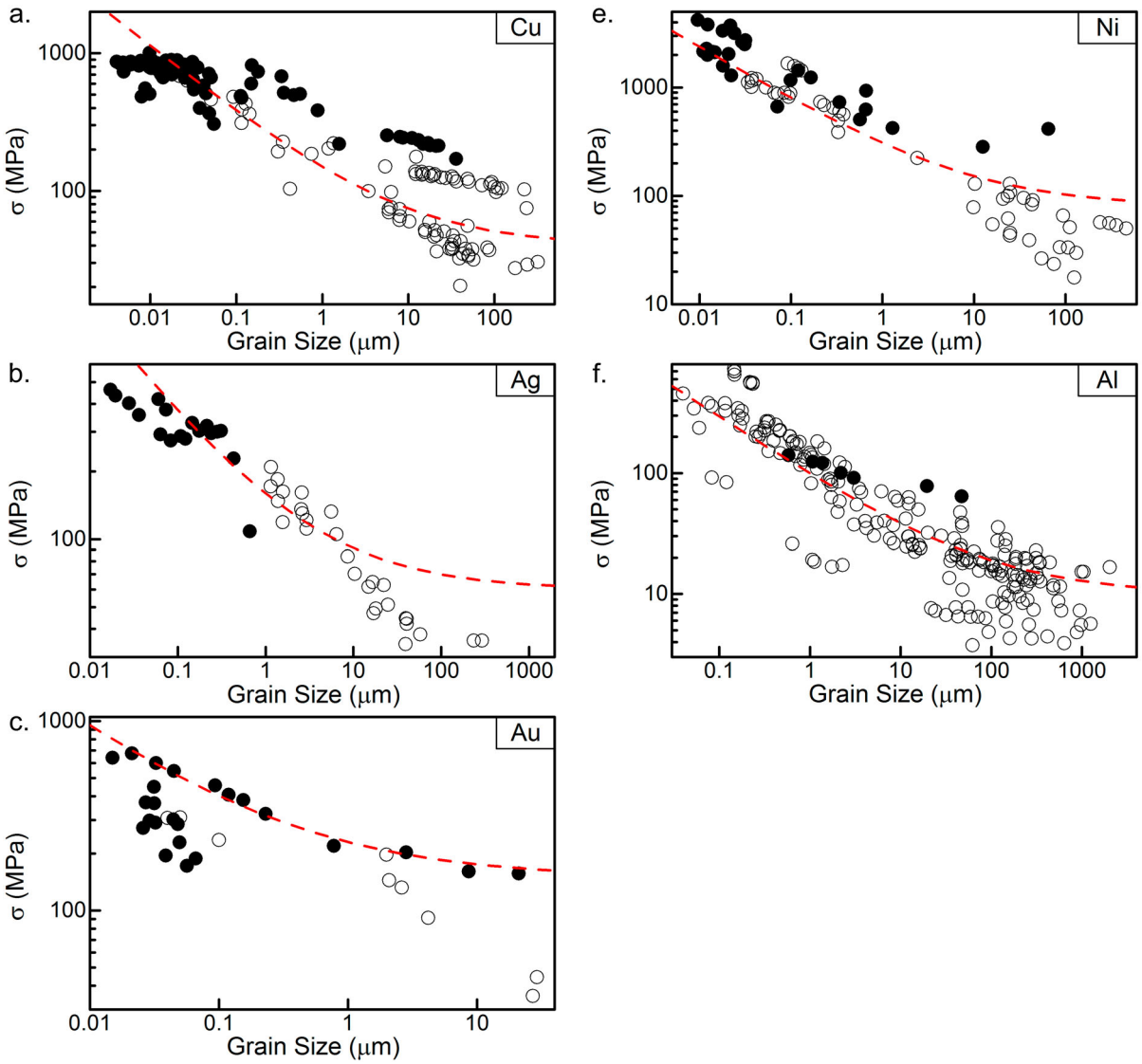
Published by Taylor & Francis on behalf of the Institute and ASM International
Received 8 December 2015; accepted 13 May 2016
DOI 10.1080/09506608.2016.1191808



1 Aggregated Hall–Petch data for each of the BCC metals as well as best fits to the data using equation (1). The closed points indicate Vickers and nanoindentation hardness measurements that were divided by a Tabor factor of 3 while the open points indicate yield strengths measured using compression or tension tests. The references included in *a*–g are as follows: *a* [12–18]; *b* [19–29]; *c* [14,30–37]; *d* [38–56]; *e* [57–60]; *f* [61–66] and *g* [67,68]

1%, because those were the smallest strain values reported. The data in Figs. 1–3 are provided in the supplementary material (Supplementary Materials 1 and 2).

One prominent feature of Figs. 1–3 is the scatter among the different studies' results. In many cases, this scatter can be ascribed to differences in sample preparation and testing methods. For example, studies on coarse-grained metals typically use specimens prepared by different thermomechanical processing techniques, e.g., swaging, rolling or repeated upset forging, followed by recrystallization anneals. These different deformation processing techniques can produce distinct textures and, depending on the environment in which they are conducted, achieve different purities, both of which can have secondary effects on the grain-size strengthening behaviour.[10,11]



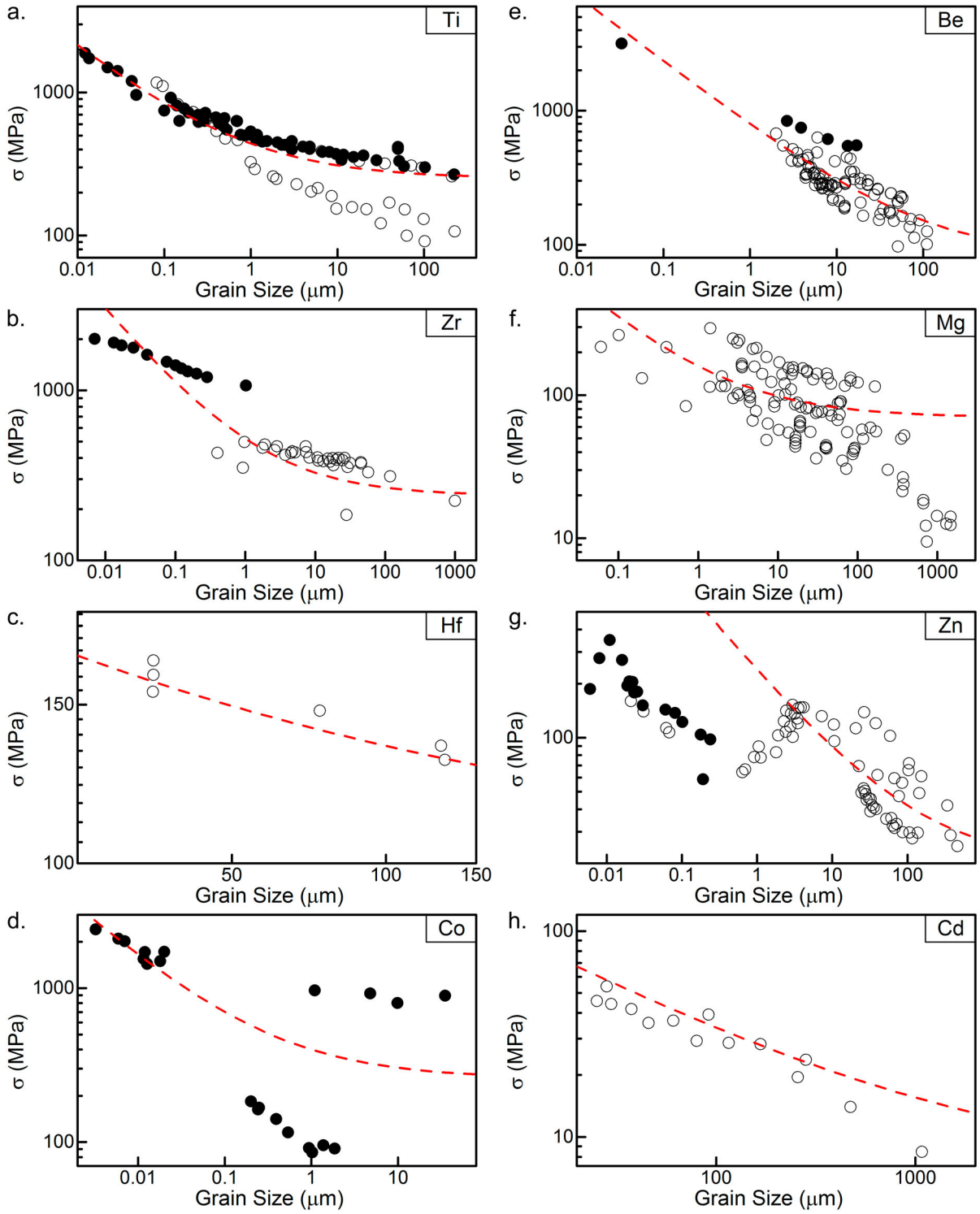
2 Aggregated Hall–Petch data for each of the FCC metals as well as best fits to the data using equation (1). The closed points indicate Vickers and nanoindentation hardness measurements that were divided by a Tabor factor of 3 while the open points indicate yield strengths measured using compression or tension tests. The references included in *a–e* are as follows: *a* [69–86]; *b* [87–91]; *c* [92–95]; *d* [96–107] and *e* [8,108–122]

Table 1 summarizes our best estimates of σ_0 and k at a plastic strain of order 0.2% for the metals included in Figs. 1–3; metals absent from Table 1 have not been studied to the best of our knowledge or, as with Pd, do not appear to have sufficient, high-quality data. The σ_0 and k values are from best fits of equation (1) to the aggregated grain-size strengthening data, with equal weighting to every data point from every study. We exercised some value judgments in this analysis, and specifically excluded data that had been questioned due to processing artefacts; this is especially the case for the finest grain sizes near the Hall–Petch breakdown where such artefacts are well-known.[86,175] The metals have been grouped by crystal structure and family in the periodic table. The range of grain sizes that have been investigated for each metal are also included in Table 1. Values of $Gb^{1/2}$ and $k/(Gb^{1/2})$, where G is the shear modulus and b is the Burgers vector, are listed in Table 1. The arithmetic mean and standard deviation of $k/(Gb^{1/2})$ for each crystal structure are shown as well.

Models for grain-size strengthening

We now turn our attention to the theoretical underpinnings of equation (1). Table 2 summarizes various different models that have been proposed to explain equation (1) and the data presented in Figs. 1–3. Most of these models treat the two terms on the right hand side of equation (1) separately: the physics associated with grain-size strengthening is assumed to be encoded in k and the grain-size exponent in the second term, while σ_0 is taken to account for all strengthening effects unrelated to the grain size.

Hall's explanation of the grain-size strengthening term in equation (1) relied on the concept of a dislocation pile-up against a grain boundary.[1] According to his model, dislocations pile-up at grain boundaries in one grain and macroscopic yielding occurs when dislocations are emitted into the adjacent grain. This is possible when the sum of the external stress and the stress at the head of the dislocation pile-up is larger than some threshold stress. The strength depends on grain size



3 Aggregated Hall–Petch data for each of the HCP metals as well as best fits to the data using equation (1). The closed points indicate Vickers and nanoindentation hardness measurements that were divided by a Tabor factor of 3 while the open points indicate yield strengths measured using compression or tension tests. The references included in *a–h* are as follows: *a* [123–132]; *b* [133–137]; *c* [138]; *d* [139–144]; *e* [6,145–156]; *f* [7,10,157–163]; *g* [164–171] and *h* [172–174]

because the total pile-up length is limited by the grain size, which therefore limits the stress at the head of the pile-up. As a result, the pile-up model predicts a linear relationship between the yield stress and the reciprocal square root of the grain size. There are many variations on this same basic model that are summarized in a review by Li and Chou,[9] but they all give a grain-size strengthening equation of the same basic form

$$\sigma = \sigma_0 + \beta G b^{1/2} D^{-1/2} \tag{2}$$

where b is the magnitude of the Burgers vector, β is a model-dependent constant that is typically on the

Table 1. Summary of results from Hall–Petch studies on pure metals

Crystal Structure	Element	Group	D_{min}/ D_{max} (μm)	σ_0 (MPa)	K (MPa μm$^{1/2}$)	Gb$^{1/2}$ (MPa μm$^{1/2}$)	k/(Gb$^{1/2}$) (–)	References
BCC	V	5	0.1/520	150	380	760	0.5	12–18
	Nb	5	0.1/610	120	340	630	0.5	19–29
	Ta	5	0.04/3900	80	760	1170	0.7	14,30–37
	Cr	6	0.02/1100	320	800	1820	0.4	57,59,60
	Mo	6	0.02/511	270	630	2070	0.3	61–66
	W	6	0.16/7	800	1000	2660	0.4	67,68
	Fe	8	0.009/2000	130	310	1290	0.2	38–44,47,49–56
						MEAN	0.42 ± 0.07	
FCC	Ni	10	0.01/460	80	230	1200	0.2	96–104,106
	Cu	11	0.005/320	40	110	770	0.1	69,70,72,73,75,76,78,85
	Ag	11	0.01/250	60	100	500	0.2	87,88
	Au	11	0.02/200	150	80	440	0.2	92
	Al	13	0.08/1200	10	90	440	0.2	8,108–122
						MEAN	0.18 ± 0.02	
HCP	Be	2	0.03/1100	80	720	680	1.1	6,145–154,156
	Mg	2	0.2/1000	70	90	100	0.9	10,157–163
	Ti	4	0.01/250	250	190	640	0.3	123–126,128–132
	Zr	4	0.4/1000	240	280	210	1.3	133–135
	Hf	4	35/130	100	350	320	1.1	138
	Co	9	0.012/4.8	260	140	380	0.4	139–144
	Zn	12	0.02/500	20	220	230	1.0	164,165,168,171
	Cd	12	28/1080	7	220	150	1.5	172–174
						MEAN	0.9 ± 0.2	

order of 0.1, and the pre-factor $\beta Gb^{1/2}$ is equivalent to k from equation (1); the values of $k/(Gb^{1/2})$ given in Table 1 correspond to β. The best fit results for the compiled data show that β is between 0.1 and 0.7 for the cubic metals, which is the same order of magnitude as predicted by the pile-up model. The data also show that β depends on crystal structure (being on average 0.18 for FCC, 0.42 for BCC and 0.9 for HCP metals) and Armstrong *et al.* demonstrated that the pile-up model can be modified to account for this.[164]

Despite these successes, the pile-up model nonetheless suffers from several major deficiencies, the most important of which is the lack of direct evidence relating the dislocation pile-up length to the grain size.[9,190] In addition, the pile-up mechanism cannot account for k's sensitivity to the average grain boundary structure and chemistry, which was demonstrated by Floreen and Westbrook in experiments on sulphur-doped nickel.[96] These issues with the pile-up model motivated the development of an alternative class of work-hardening models that assume grain boundaries influence the dislocation density, ρ, which in turn affects the flow stress through the Taylor equation:

$$\sigma = \sigma_0 + \alpha Gb\sqrt{\rho} \tag{3}$$

where α is a material-dependent constant.

The first of these work-hardening models, developed by Li, is based on the idea of grain boundary ledges that can serve as dislocation sources at yielding.[181] Li proposed that the density of these ledges scales with the grain boundary area per unit volume of material. As a result, more dislocation line length, and hence greater dislocation densities, are generated in fine-grained materials when they yield. Li showed how this mechanism can give a grain-size strengthening relationship with the same functional form as equation (1), but as with the pile-up model, there is no direct evidence linking the density of grain boundary ledges to that of dislocations.

Ashby's theory of grain-size strengthening is another work-hardening model, but it differs from the grain boundary ledge model in that it tries to explain equation (1) by reconciling two sets of observations instead of speculating about a specific mechanism. The first set of observations is that ρ increases linearly with plastic strain and inverse grain size during the uniform deformation of polycrystalline iron,[38,191] vanadium,[192] titanium,[123] niobium[29] and aluminium.[108] The second set of observations is that even when a polycrystalline specimen is subjected to macroscopically uniform plastic deformation, the individual grains still exhibit non-homogeneous plastic flow.[4,193,194] This non-uniform plastic flow is due to compatibility constraints at grain boundaries, which result in the activation of additional slip systems and the generation of excess dislocations near grain boundaries.[195] In the absence of such constraints, the individual grains in a polycrystal would simply shear on their glide planes with the largest critically resolved shear stresses.

Ashby showed that these two sets of observations are in fact related by envisioning the deformation of a tensile specimen as a two-step process.[185] In the first step, the grains undergo unconstrained, uniform deformation by shearing along their glide planes with the largest resolved shear stresses. During this first step, dislocations accumulate through random trapping events just as in a uniformly strained single crystal. The density of this group of dislocations, ρ_{SS}, is influenced by the same material and testing parameters that affect the dislocation density in a plastically strained single crystal, such as crystal structure, stacking fault energy (SFE) and homologous temperature. Since compatibility is not enforced during this first step, voids and overlaps develop between adjacent grains, and the total amount of these defects per grain is proportional to $D^2\varepsilon/2$ where ε is the macroscopic plastic strain. In the second step, these voids and overlaps are removed by introducing dislocation arrays next to the grain boundaries. The total dislocation line length that must be generated in each grain to restore compatibility

Table 2. Summary of grain-size strengthening models

Model		Year	Equation	Physics
Pile-up	Hall[1]	1951	Equation (1)	Grain boundaries (GBs) are obstacles preventing dislocation motion. Dislocations emit into adjacent grains when the shear stress at the head of a dislocation pile-up reaches some threshold value.
	Petch[2]	1953	Equation (1)	Same as above.
	Cottrell[176]	1958	$\sigma = \sigma_0 + \tau_d\sqrt{\frac{l}{D}}$ l = distance from GB to source τ_d = unpinning stress	Stresses generated by dislocation pile-ups in one grain activate Frank-Read sources in adjacent grains.
	Armstrong *et al.*[164]	1962	$\sigma = \sigma_0 + \tau_d m^2\sqrt{\frac{l}{D}}$ m = Taylor orientation factor	Similar model to Cottrell's but accounts for the fact that the dislocation source might be on a different slip system than that of the dislocation pile-up. Accounts for the effect of crystal structure on k through the Taylor factor, m.
	Smith and Worthington[177]	1964	$\sigma = \sigma_0 + \tau_d m_1 m_2\sqrt{\frac{l}{D}}$ m_1 = macro-orientation factor m_2 = micro-orientation factor	Modified version of Armstrong's model. It can account for the effect of microtexture on the activation of dislocation sources.
	Navarro and de los Rios[178]	1988	$\sigma = \sigma_0 + \frac{2\tau_d m^2}{\pi}\sqrt{\frac{l'}{D}}$ l' = 'effective position' of source	Similar model to Cottrell's but the dislocation source is assumed to extend over a finite distance.
	Nazarov[179]	1996	See reference 179, equation (30).	GB film preventing dislocation motion is broken down when the shear stress at the head of a dislocation pile-up reaches some threshold value. After yielding, dislocations accumulate in the GB and strongly affect the strength of the GB film.
	Friedman and Chrzan[180]	1998	$\sigma = \sigma_0 + \sqrt{\tau_d^2 + \frac{Gb\chi}{\pi D}}$ χ = constant	Similar to the Cottrell model but dislocations can exert backstresses that make it more difficult for Frank-Read sources to operate.
GB source	Li[181]	1963	$\sigma = \sigma_0 + \alpha Gb\sqrt{\frac{3s}{D}}$ s = line length of dislocation emitted per unit area of grain boundary α = material-dependent constant of order unity	Dislocations generated from GBs at yielding strengthen the material through the Taylor equation. The density of dislocations generated at yielding is proportional to the GB area because the dislocations are generated at GB ledges.
	Bata and Pereloma[182]	2004	See reference 187, equation (32)	Grain-size strengthening arises because of the work required to eject dislocations from GBs. Does not rely on the Taylor equation. See comments by Wert[183] and Gavriljuk[184].
GND	Ashby[185]	1970	$\sigma = \sigma_0 + CG\sqrt{\frac{b\varepsilon}{D}}$ C = material-dependent constant	Compatible deformation of individual grains in a polycrystal requires the introduction of geometrically necessary dislocations (GND's). Density of GND's is inversely proportional to the grain size. GND's affect the strength through the Taylor equation.
Composite	Thompson *et al.*[186]	1973	$\sigma = \sigma_0 + \left(1 - \frac{\lambda_S}{D}\right)\left(\frac{K_1}{D}\right) + \frac{\lambda_S}{D}\sqrt{\frac{K_2}{D}}$ λ_S = statistical slip length K_1, K_2 = constants	Treat the individual grains as composites with different dislocation densities in the grain interior and the region near the grain boundary. Dislocations affect the strength through the Taylor equation. The macroscopic flow stress is the area average stress of the two regions.
	Meyers and Ashworth[187]	1982	See reference 187, equation (33)	Elastic incompatibility leads to the generation of dislocations near the grain boundary at small strains. Treat grains at yielding as a composite with a work-hardened layer near the grain boundary.
Slip distance	Conrad[188]	1963	$\sigma = \sigma_0 + \alpha G\sqrt{\frac{b\varepsilon}{\sqrt{D}}}$ ε = strain	Assumes that the grain size affects the dislocation density, which affects the strength through the Taylor equation; that dislocations are not annihilated during the early stages of plastic straining; and that the dislocation slip distance is proportional to square root of the grain size.
	Conrad *et al.*[29]	1967	$\sigma \approx \sigma_0 + \alpha G\sqrt{\frac{b\varepsilon}{D}}$	Similar to Conrad's earlier model but assumes the slip distance is proportional to the grain size, not its square root.
	Meakin and Petch[189]	1974	$\sigma = \sigma_0 + \theta m^2 G\varepsilon + \tau_d m^2\sqrt{\frac{l}{D}} + \alpha m^{3/2} G\sqrt{\frac{b\varepsilon}{D}}$ θ = material-dependent constant	Combines the traditional pile-up model with the slip distance model due to Conrad.

is approximately $D^2\varepsilon/4b$. Because the average volume per grain is proportional to D^3, the density of this second set of dislocation is

$$\rho_{GN} \approx \varepsilon/4Db \tag{4}$$

This relationship has the same strain and grain-size dependencies seen in experiments. Note that this second set of dislocations is only introduced to ensure compatibility, so its density, to first order, is unaffected by the material and testing parameters listed above.

Ashby next showed how his model could explain the Hall–Petch equation by combining his predictions about the strain and grain-size dependence of the dislocation density with the Taylor hardening equation. Assuming for simplicity that the two groups of dislocations do not interact, the total dislocation density is $\rho = \rho_{GN} + \rho_{SS}$. Inserting this cumulative dislocation density into equation (3) gives

$$\sigma = \sigma_0 + \alpha Gb\sqrt{\rho_{GN} + \rho_{SS}} \tag{5}$$

which can be rewritten as

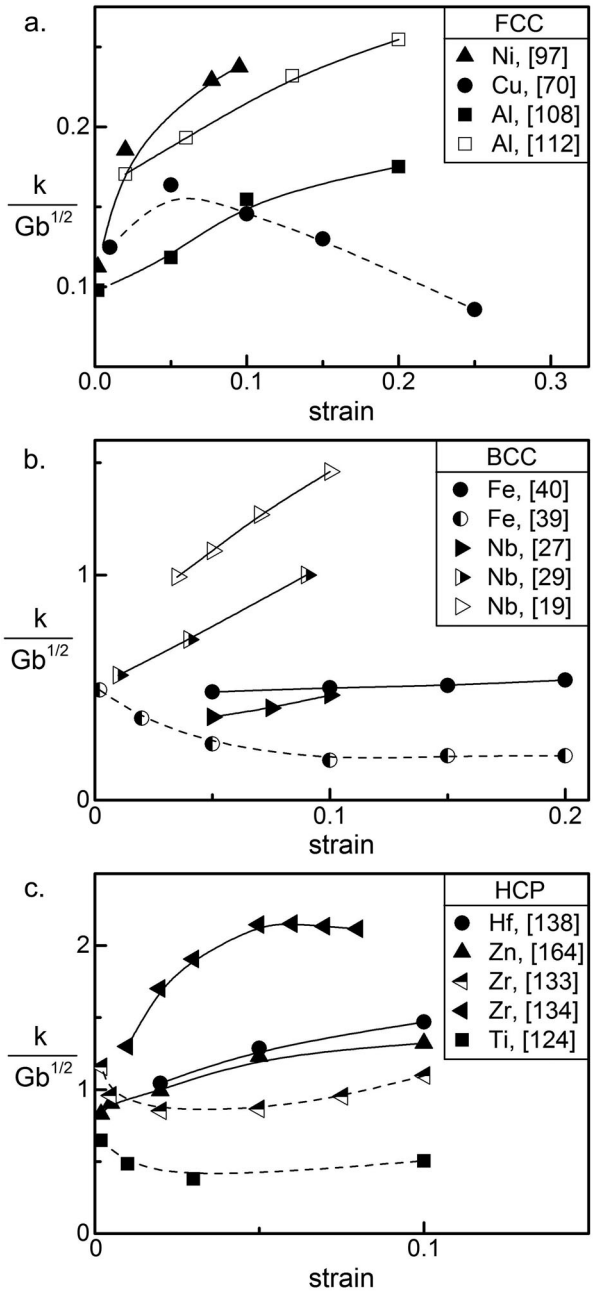
$$\sigma \approx \sigma_0 + \alpha G\sqrt{b\varepsilon/4D} \tag{6}$$

when $\rho_{GN} \gg \rho_{SS}$, as is likely the case for small plastic strains.

Comparing equations (1) and (6), we see that Ashby's model gives the correct reciprocal square root dependence on grain size, but now the Hall–Petch coefficient also depends on plastic strain, $k \propto \sqrt{\varepsilon}$. To test for this parabolic strain-dependence, several investigators have measured k as a function of plastic strain, and the results from these studies are shown in Fig. 4*a–c*. Here, the k values have been normalized by $Gb^{1/2}$ for comparison purposes.

In most of the studies shown in Figs. 4*a–c*, k increased with plastic strain, though there are some few anomalous studies in which k decreased with plastic strain. These include studies of the HCP metals titanium and zirconium,[124,133] FCC copper,[70,85] as well as BCC iron.[39] The k values of titanium and zirconium exhibit a similar trend, first decreasing and then slightly increasing with plastic strain. The most likely explanation for this initial sharp decrease in k is that these metals deform by twinning at small plastic strains and by dislocation glide at larger plastic strains. This results in an apparent decrease in k because a metal typically has a larger Hall–Petch coefficient when it deforms by twinning than when it deforms by dislocation glide, as shown by Marcinkowski and Lipsitt.[59] The strain-dependence of copper's k is the reverse of titanium's and zirconium's: its k initially increases and then decreases with plastic strain. Ono and Karashima directly linked this behaviour to the fiber-texture that can develop in recrystallized copper samples by performing tension tests on textured and texture-free samples.[70] Finally, Jago and Hansen found that iron's k decreases monotonically with plastic strain; however, these results have been contradicted by another study that showed iron's k to increase slightly with plastic strain.[39,40]

Thus, consistent with Ashby's equation (6), the Hall–Petch coefficients of most metals increase with plastic strain, and for the studies that appear to contradict Ashby's predictions, there are either physical explanations or counterexamples. Despite this qualitative agreement between experiments and Ashby's theory, none of the results in Fig. 4*a–c* exhibit the parabolic dependence on strain predicted by Ashby. Instead, non-linear fits to the data in Fig. 4*a–c* suggest that the strain exponent in equation (6) is closer to ~0.2. One possible reason for this discrepancy is that the strains at which k have historically been measured are so large that the assumption $\rho_{GN} \gg \rho_{SS}$ is no longer valid.



4 k/(Gb$^{1/2}$) as a function of plastic strain for several FCC, BCC and HCP metals

Many of the more recent models of grain-size strengthening listed in Table 2 are based on Ashby's concept of geometrically necessary dislocations.[186,187,196,197] These models treat the individual grains as composites with grain boundary and grain interior regions that accumulate dislocations at different rates and have different strengths as a result. Typically, the macroscopic yield strength in these models is the average strength of these two regions.

Finally, the mean slip distance model, proposed by several investigators,[29,188,189] arrives at a relationship similar to equation (6) and predicts many of the same behaviours as Ashby's framework, but it relies on a different set of assumptions: (1) that the average dislocation slip distance, L_{ave}, is proportional to the grain size, $L_{ave} = \eta D$, and (2) that the dislocations generated during straining do not annihilate one another. This second assumption allows the Orowan equation to be written as follows

$$\rho = \varepsilon / L_{ave} b = \varepsilon / \eta D b \tag{7}$$

Combining equations (3) and (7) gives the following grain-size strengthening relationship

$$\sigma = \sigma_0 + \alpha G \sqrt{\varepsilon b / \eta D} \tag{8}$$

with $k = \alpha G \sqrt{\varepsilon b / \eta}$, similarly to the Ashby model. The heterogeneous dislocation substructures seen in plastically deformed polycrystals seem to support the Ashby model over the mean slip distance model.[190,198]

All of the models described thus far predict a reciprocal square root dependence on grain size, in agreement with the classic Hall–Petch equation; however, several investigators have suggested that the grain-size exponent should be something other than −1/2.[188,199–204] For example, Christman fitted experimental grain-size strengthening data using a power law of the form

$$\sigma = \sigma_0 + k D^n \tag{9}$$

where n, σ_0 and k were all fitting parameters, and found that n was −1/3 for the FCC metals and varied between −1/2 and −0.9 for the BCC metals.[200] Christman did not provide a physical explanation for these alternate grain-size dependences, but other researchers have. In an early version of the mean slip distance model, for instance, Conrad proposed that the mean slip distance had a parabolic dependence on the grain size, which gives $n = -1/4$ when combined with equations (3) and (7).[188] Kocks[204] and Hirth[201] both described a composite model for which $n = -1$. And most recently, Dunstan and Bushby argued that $n = -1$ because the stress required to move a dislocation is related to its radius of curvature, which is limited by the grain size in a polycrystal.[202]

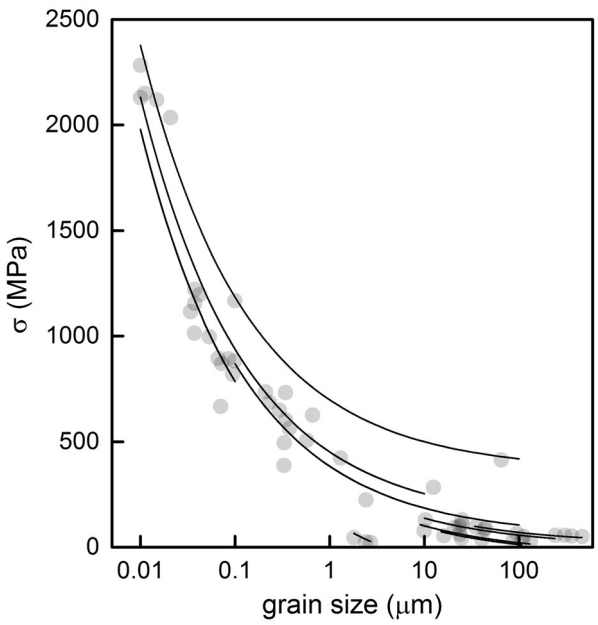
To identify the grain-size exponent that best describes the experimental results, we fitted equation (9) to each metal's aggregated grain-size strengthening data. We used a special fitting procedure to account for σ_0's sensitivity to differences in specimen preparation and characterization, which vary widely among the different studies: For a given metal, we fitted equation (9) to the results from each individual study, and treated n and k as global parameters that were same for all of the fits, and σ_0 as a local parameter that could be different for each fit. This procedure gives results like those shown in Fig. 5, which presents the grain-size strengthening data for Ni alongside the fits to the individual datasets. Each of these curves was calculated using the same values of n (−0.39) and k (330 MPa μm$^{0.39}$), but a different value of σ_0, which is why they are displaced from one another. We used this fitting procedure to calculate values of k and n for each of the metals, and the results are summarized in Table 3. Table 3 shows that while the n values range from −0.03 (Hf) to −0.95 (W), the average value is −0.40 ±0.05, which seems to support the classic Hall–Petch scaling, $n = -1/2$, over the $n = -1$ scaling that has been proposed by other investigators.[201–204] In general, the largest deviations tend to be from materials for which data are more scarce or spans a more limited range of grain sizes, with three notable exceptions – Fe, Zn and Mg. There also appear to be slight, systematic differences between the average n values for the BCC ($n_{ave} = -0.52 \pm 0.09$), FCC ($n_{ave} = -0.40 \pm 0.08$) and HCP metals ($n_{ave} = -0.29 \pm 0.07$), in line with Christman's observations. Certainly this discussion serves to highlight the need for more work examining the generality of the exponent.

The models described above explain how the grain size affects the strength, but much of the scatter in Figs. 1–3 is probably due to strengthening contributions that are not directly related to grain size. These other strengthening increments are nominally included in the frictional stress, σ_0, and below room temperature, the terms that make up σ_0 in a pure metal can be separated into two categories: temperature-dependent and athermal. The temperature-dependent component of σ_0 includes the Peierls–Nabarro stress, which is negligible for FCC and HCP metals, but can be quite large for BCC metals.[205,206] The athermal components of σ_0 include solid solution strengthening and work-hardening. We consider work-hardening as approximately athermal because the strengthening increment from a given dislocation substructure is nearly independent of temperature.

The contributions of the Peierls–Nabarro stress, solid solution strengthening and work-hardening to σ_0 are typically assumed to be independent and additive:[207]

$$\sigma_0 = \sigma_{0,PN} + \sigma_{0,SS} + \sigma_{0,WH} \tag{10}$$

Here the subscripts PN, SS and WH refer to Peierls–Nabarro, solid solution and work-hardening, respectively. In addition, each strengthening mechanism's contribution to σ_0 is approximately proportional to its contribution to



5 Aggregated grain-size strengthening data for Ni as well as best fits using equation (9) to the results from the individual studies

the critical resolved shear stress of a single crystal:

$$\begin{aligned} \sigma_{0,PN} &\approx M_{\text{ave}}\,\tau_{0,PN} \\ \sigma_{0,SS} &\approx M_{\text{ave}}\,\tau_{0,SS} \\ \sigma_{0,WH} &\approx M_{\text{ave}}\,\tau_{0,WH} \end{aligned} \qquad (11\text{a}-\text{c})$$

where $\tau_{0,PN}$, $\tau_{0,SS}$ and $\tau_{0,WH}$ are the respective strengthening increments in a single crystal from the Peierls–Nabarro stress, solid solution strengthening and work-hardening, and M_{ave} is the Taylor factor.[41,42,59,108,164] The Taylor factor accounts for the homogeneous deformation of the individual grains in a polycrystal.[208]

We can evaluate the validity of equations (11a–c) by comparing the temperature and strain dependencies of σ_0 with those of single crystal yield and flow stresses. For instance, Fig. 6a, taken from the work by Marcinkowski and Lipsitt,[59] shows the σ_0 and $M_{\text{ave}}\tau_0$ of Fe as a function of temperature at constant strain, and illustrates how the datasets are very similar in magnitude and form, in line with equation (11a). In Fig. 6*b–e*, we repeat the Marcinkowski–Lipsitt analysis for several other BCC metals, plotting the σ_0 and $M_{\text{ave}}\tau_0$ as a function of temperature at constant strain, and demonstrate that equation (11a) applies to these other metals as well. The outlying points in Fig. 6 can be explained when we account for differences in sample purity and the contribution of solid solution strengthening to σ_0. For example, in Fig. 6*d*, solid solution strengthening due to the large C concentration (1000 ppm) in Lindley and Smallman's vanadium specimens resulted in the 250 MPa offset of their results relative to the other measurements.[12]

Figure 6 only shows data for BCC metals, as the FCC and HCP metals do not have nearly as significant a volume of data to examine. Nonetheless, FCC metals tend to exhibit the same trends, except the effect of temperature is much less dramatic than it is for BCC metals, which is expected given FCC metals' generally weaker Peierls–Nabarro stresses. On average, the σ_0 and $M_{\text{ave}}\tau_0$ of FCC metals increase at most just a few dozen MPa on cooling from room temperature to 77 K.[70,71,97,109] The HCP metals behave similarly to the FCC metals, aside from anomalous reports of large increases in Zr's σ_0 on cooling by Coleman and Hardie[133] and Ramani and Rodriguez,[134] which may have been due to the onset of deformation twinning.

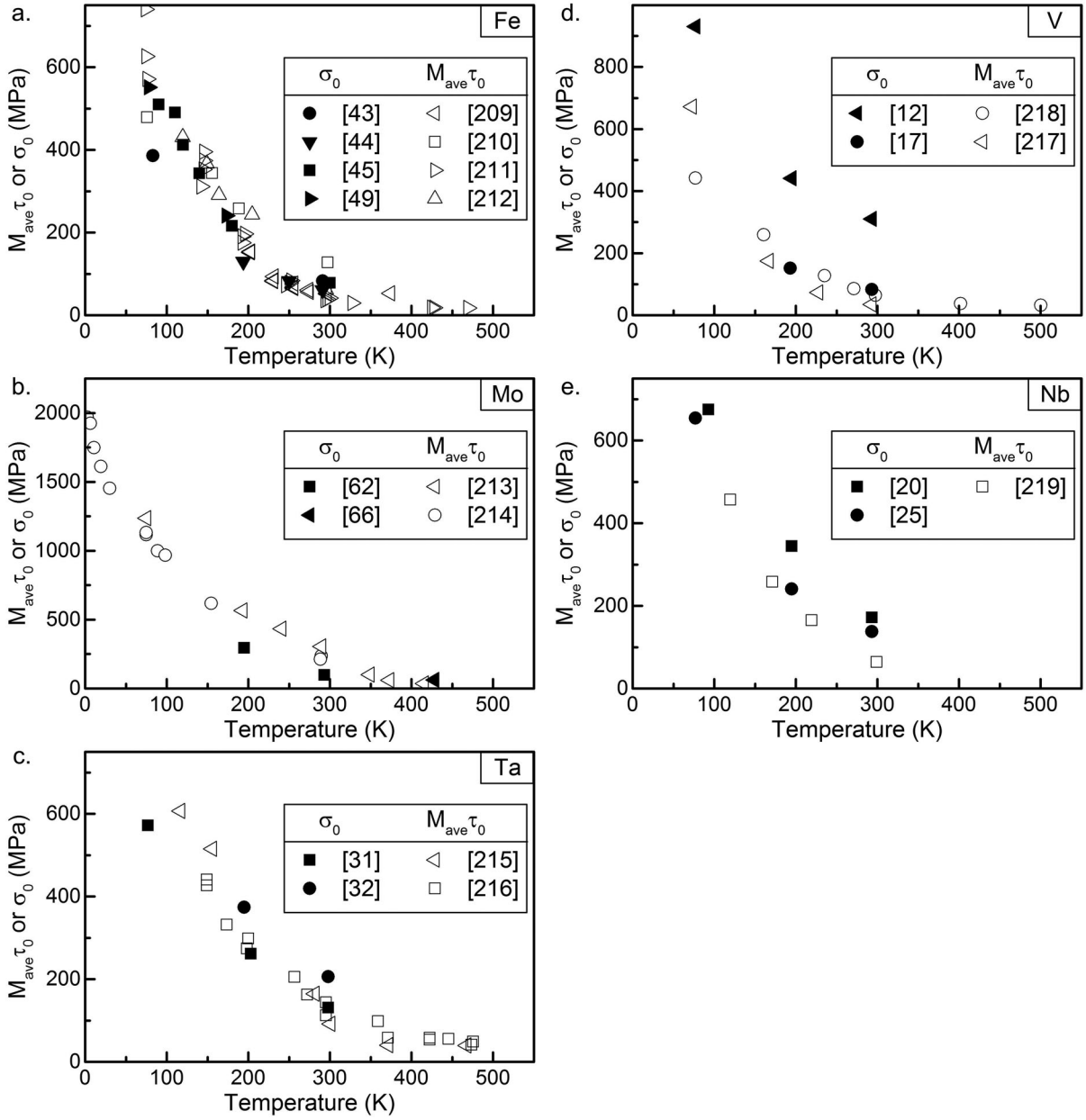
Following Hansen,[108] we illustrate the contribution of work-hardening to σ_0 by plotting plastic strain against σ_0/G, where G is the shear modulus, in Fig. 7*a–c*. In line with equation (11c), the increase in σ_0/G with plastic strain seen in these figures resembles the increase in τ_0/G due to work-hardening during the axisymmetric deformation of single crystals.[204] This effect of plastic strain on σ_0/G was first observed in early Hall–Petch studies;[71,108] however, in the context of the present larger dataset, new trends in the plastic strain-dependence of σ_0/G become apparent. For example, closer inspection of Fig. 7*a–c* reveals that material and testing parameters affect work-hardening's contribution to σ_0/G in the same way they influence its contribution to τ_0/G. Comparison of Fig. 7*a* and 7*b* shows that σ_0/G increases more rapidly with plastic strain in FCC metals than it does in BCC metals. This is consistent with FCC single crystals' much greater normalized rates of work-hardening than BCC or HCP single crystals'.[220] Figure 7*b* shows that FCC metals with smaller stacking fault energies exhibit greater increases in σ_0/G at a given strain. Comparing the results from tests performed at 77 K shown in Fig. 7c with those in Fig. 7b demonstrates that plastically straining at lower testing temperatures can result in larger increases in σ_0/G at a given strain. These last two trends are evident in tests on FCC single crystals as well, where lower stacking fault energies and testing temperatures are known to increase the rate of work-hardening by suppressing dynamic recovery.[220,221]

The similar strain dependencies of σ_0/G and τ_0/G support Ashby's argument that dislocations can be divided into two groups. One group consists of the statistically stored dislocations that contribute to changes in σ_0/G. The density of these dislocations at a given strain is comparable to the dislocation density of a single crystal subjected to the same amount of uniform deformation and is strongly affected by variables such as crystal structure, SFE and temperature (cf. Fig. 6). The second group of dislocations contains the excess dislocations that ensure the compatible deformation of the individual grains in a polycrystal. And in the context of the Ashby model, it is this second group that gives rise to grain-size strengthening.

Returning now to the aggregated grain-size strengthening results shown earlier, we can use the concepts just introduced to explain much of the scatter seen in this data. As an example, consider Fig. 8 which shows fits using equation (1) to the niobium yield stress data from references [19,20,23,24,27,29]. The large offsets between the fits in this figure are primarily due to solid solution strengthening from interstitial contaminants whose concentration varied from study to study. To demonstrate this, in Table 4 we compare each study's σ_0 with the expected solid solution strengthening increment due to their specimens' oxygen and nitrogen contents, σ_{SS}. As Table 4 shows, the samples from reference [27] had the smallest solid solution strengthening increment, which is why their σ_0 is the smallest and their results sit well

Table 3. *n* and *k* values from best fits using equation (9) to each metal's aggregated grain-size strengthening data

	n	*k* (MPa μm^{-n})
V	−0.47	280
Nb	−0.54	500
Ta	−0.54	260
Cr	−0.58	650
Mo	−0.32	1420
W	−0.95	410
Fe	−0.29	640
Ni	−0.39	330
Cu	−0.38	160
Ag	−0.17	300
Au	−0.47	100
Al	−0.60	100
Be	−0.56	870
Mg	−0.21	150
Ti	−0.47	210
Zr	−0.19	540
Hf	−0.03	800
Co	−0.51	100
Zn	−0.15	80
Cd	−0.17	190

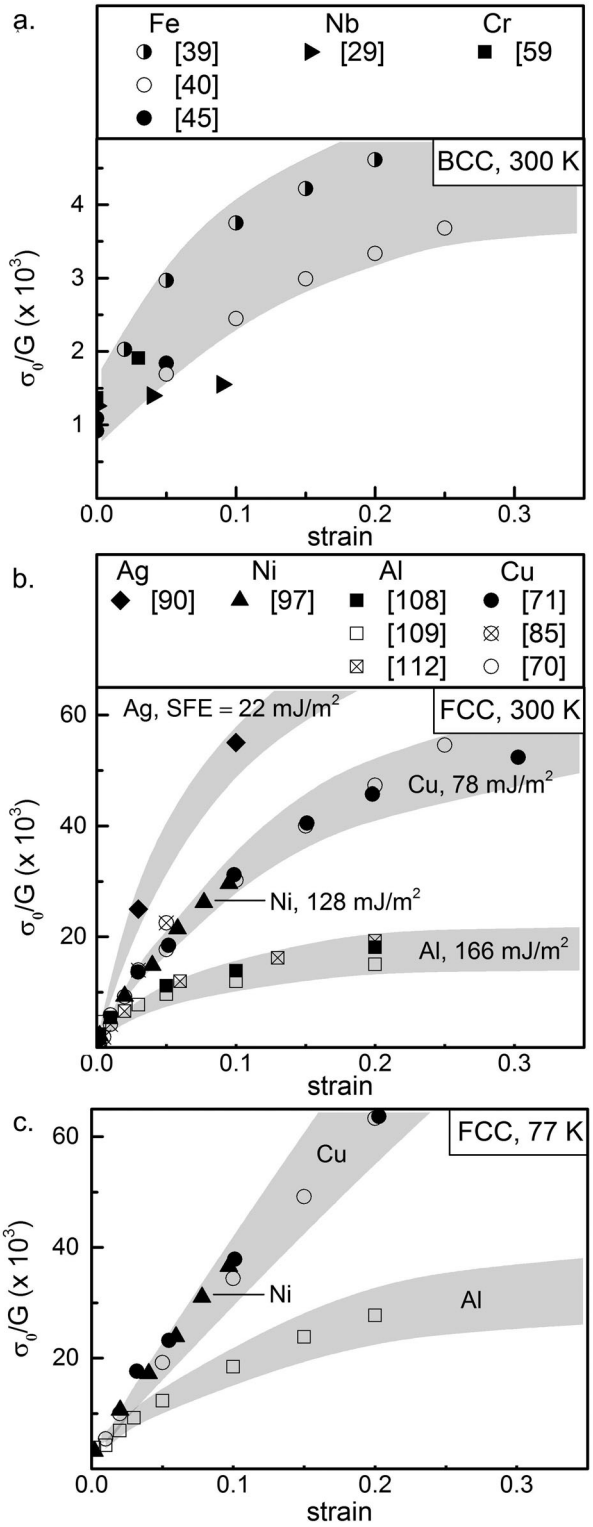


6 Testing temperature versus $M_{ave}\tau_0$ from tension tests on single crystals and σ_0 from fits to yield stress measurements using the Hall–Petch equation. We used $M_{ave} = 2.7$ which is appropriate for BCC polycrystals that exhibit pencil glide and have no texture.[208] The close correspondence between $M_{ave}\tau_0$ and σ_0 supports equation (11a). The large and rapid increase in both $M_{ave}\tau_0$ and σ_0 with decreasing temperature reflects the BCC metals' large, temperature-sensitive Peierls–Nabarro stress. The single crystal data in *a–e* are from the following references: *a* [209–212]; *b* [213,214]; *c* [215,216]; *d* [217,218] and *e* [219]

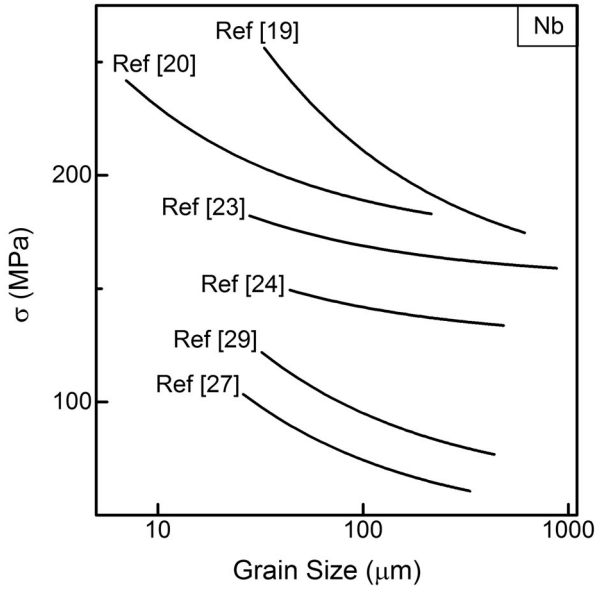
below the others in Fig. 8. By comparison, the other studies used relatively impure samples, and the values in Table 4 show that for several of these other studies, their predicted solid solution strengthening increments can account for the difference between their σ_0 and that of reference [27].

In the supplementary material, we provide a summary of the experimental details from each of the Hall–Petch studies cited herein. This table includes the specimen purity, the processing techniques used to prepare the test specimens, the mechanical test method, and the technique used to measure the grain size. This information can be used to explain some of the differences in σ_0 among the various studies on a given metal.

Work-hardening is not important in Fig. 8 because the niobium results were all measured at the same plastic strain, but it does contribute to the scatter in chromium's grain-size strengthening data shown in Fig. 9. Here the hardness measurements from reference [60] are offset ~200 MPa above the yield strength measurements from reference [59], and this is because hardness measurements include some plastic strain. Following Tabor, the Vickers hardness is equal to the flow stress of a test specimen after it has been plastically strained an additional 8%.[223] Work-hardening due to this additional 8% plastic strain can account for the ~200 MPa difference between the hardness and yield strength data, as shown by Brittain *et al.*[60] Such differences between hardness and yield strength measurements are seen in almost every metal's grain-size strengthening data. The effect is most evident among studies on coarse-grained metals because they



7 *a* Increase in several BCC metals' σ_0/G with plastic strain at room temperature. *b*, *c* Increase in several FCC metals' σ_0/G with plastic strain at room temperature and 77 K, respectively. Note the close resemblance of these results to the increase in the flow stress of single crystal test specimens oriented for multislip during tension or compression testing. The material and testing parameters affecting the normalized rate of work-hardening of single crystals – crystal structure, SFE,[222] temperature – have the same effect on σ_0/G's strain-dependence



8 Independent fits using equation (1) to the niobium grain-size strengthening data from references [19,20,23,24,27,29]

work harden to a greater extent than ultrafine grain or nanocrystalline metals.

The Hall–Petch breakdown

For coarse-grained specimens, the preceding discussion demonstrates that the grain-size strengthening increment can be estimated using equation (1) and the constants in Table 1, and that much of the variation among the different studies can be explained by some combination of specimen contamination and work-hardening effects. However, data from a few of the datasets in Figs. 1–3 (most notably that for Zn and Cu) illustrate that simply extrapolating the grain-size strengthening behaviour of coarse-grained samples to nanocrystalline grain sizes can result in errors: the strength measured in the finest grain-size specimens can be less than that expected.[225,226] This apparent breakdown of the Hall–Petch effect has been much discussed and debated, and it is clear that many of the observations of lowered strength levels in nanograined materials can be attributed to artefacts in sample processing. At the same time, it is physically reasonable that the Hall–Petch equation breaks down;

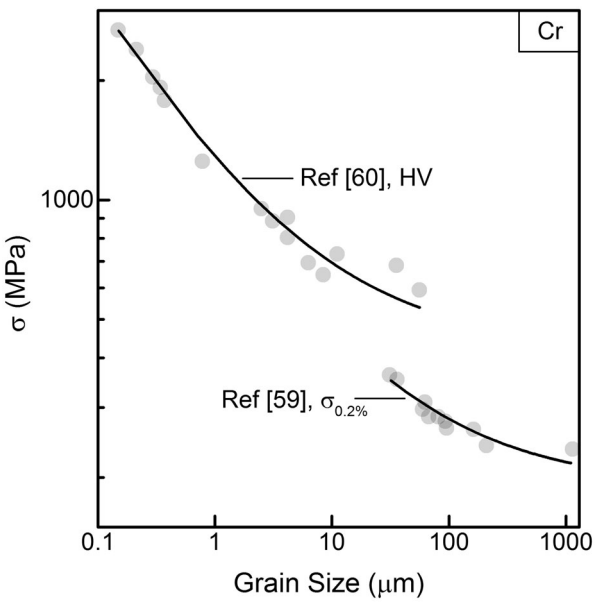
Table 4. Interstitial content and corresponding solid solution strengthening increment in the Nb studies shown in Figure 7. The solid solution strengthening increments for oxygen and nitrogen are 0.19 and 0.41 MPa/ppm, respectively[224]

References	ppm by wt O	*N*	$\sigma_{0,ss}$ (MPa)	σ_0 (MPa)
27	120	13	30	40
29	60	45	30	60
24	200	200	120	130
19	550	120	160	150
20	400	100	120	170
23	185	57	60	150

for instance, if it did not, the yield strength of a tungsten specimen with a grain size of 10 nm would be larger than tungsten's theoretical yield strength, $G/15 = 10$ GPa.[227]

In-situ and ex-situ TEM investigations of plastic deformation in nanocrystalline nickel,[228,229] copper,[230,231] gold,[232] aluminium[233] and platinum[234] show a transition in the dislocation behaviour in very fine-grained samples. These studies have shown that while there can be extensive dislocation activity in samples with grain sizes as small as 10 nm, dislocations do not accumulate in nanocrystalline metals as they do in coarse-grained samples. Instead, in nanocrystalline metals, dislocations emitted from one grain boundary are often rapidly absorbed by other grain boundaries after they traverse the grain interior. Measurements showing a plateau or even a decrease in the dislocation density at ~100 nm suggest that this transition in the dislocation behaviour may also contribute to a lowering of the Hall–Petch slope at such grain sizes.[235,236] In light of these results, it seems that the strength of nanocrystalline metals is determined by the stress required to nucleate, propagate or reabsorb individual dislocations, and this motivated the dislocation-nucleation models of the Hall–Petch breakdown developed by Asaro and co-workers,[237–240] which are relevant in the range of grain sizes down to perhaps ~10 nm.

At the finest grain' sizes there is considerable evidence that the Hall–Petch breakdown involves a weakening effect as grain size is reduced beyond about 10–20 nm. The effect is most thoroughly experimentally established on alloy systems in which such fine grains can be stabilized, and experimental data on pure metals with grain sizes in this range are rare and not usually systematic. It is the authors' opinion that conclusively demonstrating the Hall–Petch breakdown by experiments on a pure metal will be difficult or even impossible, because of the rapid grain growth in pure nanocrystalline metals at low homologous temperatures, the possibility of grain growth during plastic deformation and the fact that grain boundary structure and state can affect strength to the same extent as grain size.[48] There are, however, several systematic computational studies on pure metals that convincingly demonstrate the inflection of strength at a 'strongest grain size.'[241–244] The strength inflection at these fine grain sizes is associated with a transition to intergranular deformation mechanisms dominating flow. Reference [245] argues that these mechanisms are very similar to those responsible for deformation in metallic glasses, and shows that many unusual metal properties emerge at the finest grain sizes that are consistent with such a picture: a loss of rate sensitivity in FCC metals, the emergence of pressure sensitivity and a tendency for shear localization are three features of glass plasticity that support the Hall–Petch breakdown as being reflective of a cross-over to glass-like deformation physics in the intergranular regions.



9 Chromium grain-size strengthening data from references [59,60] as well as best fits using equation (1) to each study's data

Many of the other models that have been developed to explain the Hall–Petch breakdown are modified versions of those originally developed to explain equation (1). For example, one class of models, based on the pile-up concept, assume that the Hall–Petch breakdown occurs because either the grains in the nanocrystalline metals are too small to support a dislocation pile-up[179,246,247] or the grain size affects the pile-up's interactions with the grain boundaries.[248] There are also modified composite models in which the bulk material's strength is assumed to be the average of a weaker grain boundary phase and a stronger grain interior phase,[249–257] which is physically in line with the transition to intergranular 'glass-like' plasticity described above. More detailed composite models account for additional microstructural features including triple junctions and quadruple nodes.[258–261] One hypothesized reason the grain boundaries appear weaker than the grain interior is that they accelerate Coble creep processes;[86,259,262,263] however, because these Coble creep models require strain rate sensitivities that are inconsistent with measured values, their validity has been questioned.[264,265]

Summary and outlook

In this work, we calculated best estimates of the pure metals' Hall–Petch parameters using the grain-size strengthening data accumulated over the past six decades. We also used these data to validate the Hall–Petch coefficient's dependence on crystal structure, the similarities between the frictional stress and single crystal flow stress and several other trends. Some of these trends are accounted for in the dislocation pile-up model of grain-size strengthening, but as we have noted, there is no direct evidence to support this model's underlying assumption that the grain-size limits the size of dislocation pile-ups. Though the other explanations of grain-size strengthening that we summarized also have deficiencies, Ashby's geometrically necessary dislocation model appears to be the most consistent with the strain-dependence of the Hall–Petch coefficient and with experimental observations of the dislocation substructure in plastically deformed polycrystals with coarse grain sizes.

Our compilation of grain-size strengthening measurements revealed several gaps in the open literature that should be addressed in future research. In addition to the obvious gaps related to, e.g., materials that have not yet been studied, there is a dearth of accurate strength measurements on samples with sub-micron grain sizes,

which is in part due to the difficulty of preparing bulk, defect-free, nanocrystalline specimens for standard tension and compression testing. Recently, many of these processing challenges have been overcome,[266,267] and these advances in processing should enable more systematic investigations of grain-size strengthening in ultrafine and nanocrystalline metals.

Future research on grain-size strengthening should also address the relationship between grain-size strengthening and grain boundary structure. This includes detailed work on grain boundary crystallography and the sampling of boundary crystallographic space in a polycrystal, as well as details associated with the thermodynamics of interface structure, i.e., grain boundary complexion.[268] While these topics have been actively studied in the grain boundary community for many years, the connection between boundary structure and grain-size scaling behaviour remains fertile territory for future work. Such connections would be especially relevant in materials with very fine grain sizes, where the grain boundaries themselves can accommodate plastic strain.

Insight into the relationship between grain boundary structure and strength may also come from studies on nanotwinned metals, which contain large populations of coherent twin boundaries with much smaller excess energies and free volumes than the high angle boundaries typically found in traditional nanocrystalline metals.[222] Because nanotwinned and traditional nanocrystalline metals have such disparate grain boundary structures, differences in their mechanical properties affected by grain boundary structure are maximized, and this makes such properties easier to identify. For example, initial studies comparing nanotwinned and nanocrystalline copper found that nanotwinned copper's yield strength exhibits a Hall–Petch type dependence on the mean twin spacing with a frictional stress and Hall–Petch coefficient similar to those of nanocrystalline copper, but that nanotwinned copper work-hardens faster than traditional nanocrystalline copper does.[269,270] These results indicate that the effects of grain boundary structure on mechanical properties can persist to large plastic strains, which is an unexpected finding that should be further clarified.

Another topic that should be investigated in more detail is the grain-size dependence of the yield stress of materials that mechanically twin. Several studies on BCC metals have shown that even when these materials twin, their yield stress is still related to the grain size through equation (1).[12,59] These studies have also established that, at a given temperature, the Hall–Petch coefficient for twinning can be up to an order of magnitude greater than the Hall–Petch coefficient for dislocation glide.[271] Explanations of these behaviours have been proposed that are based on the dislocation pile-up concept[272] and the grain-size dependence of the twin morphology,[129] but there is limited direct experimental validation of these theories. Future work on this topic will be of engineering value because it may help guide the design of microstructures that are more resistant to twinning and are therefore more ductile.

Though the Hall–Petch equation has been investigated for over half a century – and grain-size strengthening for even longer – there are still many exciting and relatively unexplored avenues for future research. Work on these topics will benefit from recent developments in imaging, mechanical characterization and processing, and any advances in understanding that are made through this work will have a significant impact on materials engineering, since they will enable ever finer control over the mechanical properties of metals.

Acknowledgements

The authors acknowledge the partial support of several agencies in the development of this review, including the U.S. Army Research Office under grant W911NF-14-1-0539 and through the Institute for Soldier Nanotechnologies at MIT, as well as the U.S. National Science Foundation under grant CMMI-1332789.

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